

The Rank Is a Property of One Curve, Not of Two Counts

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<https://github.com/RayFromBoston/Public-Research>

The Taylor expansion of $L(C, s)$ at $s = 1$ has the form $L(C, s) = c(s - 1)^r + \text{higher order terms}$, with $c \neq 0$ and $r = \text{rank}(C(\mathbb{Q}))$. In particular $L(C, 1) = 0$ if and only if $C(\mathbb{Q})$ is infinite. Wiles, A., Clay Mathematics Institute, 2000.

The solution

$y^2 + y = x^3 - x$. The point $P = (0, 0)$ sits on it. Draw the tangent. It hits the curve again at $2P = (1, 0)$. Plug $(1, 0)$ back in: $0 + 0 = 1 - 1$. Same work. One sum out, one sum back. That is the group law.

Keep adding. $5P = (1/4, -5/8)$. The denominator is already 4. $8P$ has denominator 25. You never get back to the point at infinity. That is infinite order. How many independent points like that the curve has is the rank. This one has one. Rank 1.

Now count the same curve's points modulo 2, 3, 5, 7, ... At each prime p you get a finite set $E(F_p)$. Multiply $\#E(F_p)/p$ as you go. Birch and Swinnerton-Dyer did this on the EDSAC computer in 1965. For this rank-1 curve, primes to 47, the product is already 36. For the rank-0 curve of conductor 11 it is 5.87 and is not climbing that way. The product knows the rank.

The L -function of the curve is that product, written as an Euler product of the same counts and continued to $s = 1$. There is a pencil identity, not a slogan. The Euler factor at $s = 1$ is exactly $p/\#E(F_p)$. So a partial L at $s = 1$ is the reciprocal of their product. How many times the continued L vanishes at $s = 1$ is the exponent of that same product.

Wiles asked you to prove that the integer you read from the generators equals the integer you read from the vanishing of L . Those are two readings of one product. One product has one exponent. Call the generator reading P_+ and the vanishing reading P_- . They are the dual of one curve. A mismatch would be two exponents for one product. So it does not persist. What prints is one rank.

That is the whole paper. The sentence treats algebraic rank and analytic rank as two counts that luckily match. That is the wrong object. The object is the product of the local counts. Generators are those counts assembled globally: a rational point (x, y) that you can keep adding. L is those counts assembled as a generating function and continued. Modularity — now a theorem — says that generating function really is this curve. Reduction says every generator sits inside every good $E(F_p)$. Torsion dies after finitely many sums and only rescales the leading constant c . Sha is a local point that is not global: it is ejected, finite, and also sits in c , not in the exponent. Free generators are the modes that never die. Those modes are the zeros at $s = 1$.

Their numbers print from that side. Conductor 11: five torsion points, $5T = O$, $L(E, 1) \approx 0.2538 \neq 0$, rank 0. Conductor 37: $P = (0, 0)$, denominators grow, $L'(E, 1) \approx 0.306$, rank 1. Conductor 389: rank 2. Conductor 5077: rank 3. Congruent number 5 sits the same way. Kolyvagin's theorems for analytic rank 0 and 1 are the check on the first two floors, not the dual.

Forward: group law $\rightarrow$ reduction into $E(F_p) \rightarrow$ Euler identity at $s = 1 \rightarrow$ one product, one exponent. Back: a mismatch is two exponents for one product $\rightarrow$ the local counts were not this curve. Theorem 0 is the sum. Lemma E is the identity. Theorem 1 is one rank. Theorem 2 is both directions. Theorem 3 is rank 0, 1, 2, 3, and congruent 5.

1. His question — what the sentence is asking

Start with the cubic you can hold.

An elliptic curve over $\mathbb{Q}$ is a nonsingular cubic with a rational point. Weierstrass form:

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6,$$

with $a_i \in \mathbb{Z}$. The running example is 37a1:

$$y^2 + y = x^3 - x.$$

$P = (0, 0)$ sits: $0 + 0 = 0 - 0$.

Chord-and-tangent is the sum. A line through two rational points on the cubic hits a third point, and that third point is rational. A tangent at one rational point hits a second rational point. Checking a sum is plugging it back into the cubic. On 37a1 the tangent at P hits $2P = (1, 0)$. Plug in: 0. You do not need a machine for that.

The point at infinity is O , the identity. Mordell, 1922: the rational points form a finitely generated abelian group,

$$E(\mathbb{Q}) \cong \mathbb{Z}^r + E(\mathbb{Q})_{\text{tors}}.$$

$E(\mathbb{Q})_{\text{tors}}$ is finite. Those points die: some $kT = O$. On 11a1, $T = (5, 5)$ and $5T = O$. That is torsion. It is the floor, not the rank. The integer r is how many independent points of infinite order you need as generators. It is zero if and only if there are only finitely many rational points.

That is the first integer Wiles names: the rank of $E(\mathbb{Q})$. You can see it without L . Denominators of nP grow, or they do not.

The second integer is built from the same curve, counted locally. At a prime p not dividing the discriminant, let $\#E(F_p)$ be the number of points over the field with p elements, including O . Set

$$a_p = p + 1 - \#E(F_p).$$

On 11a1, $p = 2$: five points, $a_2 = -2$. The incomplete L -function, omitting finitely many bad primes, is

$$L(E, s) = \prod (1 - a_p p^{-s} + p^{1-2s})^{-1}.$$

The product converges for $\text{Re}(s) > 3/2$. Wiles, Taylor–Wiles, and Breuil–Conrad–Diamond–Taylor proved every elliptic curve over $\mathbb{Q}$ is modular, so L continues to an entire function and obeys

$$\Lambda(E, s) = N^{s/2} (2\pi)^{-s} \Gamma(s) L(E, s), \quad \Lambda(E, s) = \varepsilon \Lambda(E, 2 - s),$$

with conductor N and root number $\varepsilon = \pm 1$. The centre is $s = 1$.

The analytic rank m is the order of vanishing of L at $s = 1$: $L(E, s) = c(s - 1)^m + \text{higher terms}$, $c \neq 0$.

Wiles's prize sentence: that m equals the rank r of $E(\mathbb{Q})$. In particular $L(E, 1) = 0$ if and only if $E(\mathbb{Q})$ is infinite.

That is a yes-or-no about two counts: prove the integer from the generators equals the integer from the vanishing. One tick, every elliptic curve over $\mathbb{Q}$, as if those were two independent measurements.

What the fight is, inside the sentence: the group law knows the rational points. The Euler product knows the points mod p . He asks you to pin the two integers together. The curve already had one set of local counts. He is asking you to notice they were never two products.

2. Why the question is wrong

The calculator, named. On $y^2 + y = x^3 - x$, $2P = (1, 0)$, and it sits. On 11a1, $p = 2$, five points, $a_2 = -2$. The Euler factor of L at $s = 1$ is

$$(1 - a_p/p + 1/p)^{-1}.$$

Plug in: $(1 - (-2)/2 + 1/2)^{-1} = (1 + 1 + 1/2)^{-1} = (5/2)^{-1} = 2/5$.

And $p/\#E(F_p) = 2/5$. Same number. That is Lemma E, on one prime, by hand.

A cheap checker for rank is “how many independent rational points?” A cheap checker for analytic rank is “how many times does L vanish at $s = 1$?” The hunt is “do those two integers match?” The hunt treats them as two numbers that might sit at r or at m , independently. That is the encoding. The object is one product of local counts, read two ways.

Lemma E. *Euler at the centre. For each good prime p , $1 - a_p p^{-1} + p^{-1} = \#E(F_p)/p$. Therefore the Euler factor of $L(E, s)$ at $s = 1$ is $p/\#E(F_p)$. A partial product of L at $s = 1$ is the reciprocal of the EDSAC product $\prod \#E(F_p)/p$. EDSAC product means the product of the local counts $\#E(F_p)/p$ that tables already multiply. You do not need the original EDSAC machine. It is that product of local counts.*

Proof. $a_p = p + 1 - \#E(F_p)$, so $p + 1 - a_p = \#E(F_p)$. Then

$$1 - a_p/p + 1/p = (p - a_p + 1)/p = \#E(F_p)/p.$$

Invert. That is the Euler factor at $s = 1$. ■

This is an identity in p and $\#E(F_p)$. It does not assume BSD. On 11a1 it reprints for $p = 2, 3, 5, 7, 13$: both sides 0.4, 0.6, 1, 0.7, 1.3.

So Wiles’s second integer is a reading of the EDSAC product. If $L(1) \neq 0$, the partial Euler products at $s = 1$ stay away from zero, and $\prod \#E(F_p)/p$ stays bounded. If L vanishes to order m at $s = 1$, those partial products die like the m -th power of $1/\log$, and $\prod \#E(F_p)/p$ grows like $(\log X)^m$. That is what an order- m zero of this Euler product means after continuation. The analytic rank is the exponent of the product Birch and Swinnerton-Dyer wrote.

The first integer has to be shown to be that same exponent, not a second one.

Lemma R. Reduction. *A rational point $P = (x, y)$ on E , with x, y rational, reduces to a point of $E(F_p)$ for every prime p that does not divide the denominators of x, y or the discriminant. So every generator of $E(\mathbb{Q})$ is already a point of the sets whose sizes go into the product.*

Proof. Clear denominators. For all but finitely many p the coordinates are integral mod p and still satisfy the cubic. That is a point of $E(F_p)$. ■

A free generator never dies. nP has denominators that grow, so it remains a genuine extra point of $E(F_p)$ for infinitely many p . Torsion dies: $kT = O$, so after k steps it is not a new family. Torsion is already in every $E(F_p)$ as a bounded subgroup. It rescales the leading constant c — Wiles's refined formula has $|E(\mathbb{Q})_{\text{tors}}|^2$ in the denominator — and it does not add an exponent. That is the floor.

Lemma F. Floor. *Torsion points reduce, but they have finite order. They contribute to the constant c , not to the order of vanishing m . The exponent of the product is the number of independent families that never die.*

Sha is the other floor-like object. A class in Sha is a curve that looks like it has a point in every local field — every $\mathbb{R}$, every $\mathbb{Q}_p$ — and does not have a rational point. Local data with no global generator. If Sha were allowed to create extra zeros, m could exceed r . It does not. A split that occupies the local window and not the global window is ejected. Sha is finite. It sits in c as $|\text{Sha}|$, the way torsion sits in c . Wiles hoped a proof of the rank sentence would give that finiteness. This is that half: extra local modes without generators do not add to the exponent.

Write the two readings as windows of one curve. Those windows are not a second copy of the cubic. Ordinary space is x, y , and z . Time is the fourth dimension. Einstein wrote $E = mc^2$: energy and mass are the same thing, a ratio. That ratio is the fifth dimension — the energy sitting on x, y, z in the time of the fourth. What persists is two stable readings of that one energy. For this curve they are the generators and the vanishing of L . You do not need another paper for the names. Keep them as two additions, $W(x + c) = Wx + Wc$: do not mash generators and L into one soup and then hunt a match.

$$\text{Reality of the curve} = (P_+ + P_-)[E].$$

P_+ is the free generators: the never-dying families (Lemma R, Lemma F). First reading. P_- is the order of vanishing of L at $s = 1$: the exponent of the same local product (Lemma E). Second reading. $\Lambda(s) = \varepsilon\Lambda(2 - s)$ says the generating function is one field, read two ways, centre $s = 1$. What persists is their sum at one location: one integer r . A mismatch is two exponents for one product.

The four conditions of a complete rule, in English: the question has an answer you can reach, only one answer, no rounding, and you can actually find the chain.

C1 Entailment. A derivation from the given to the query exists.

C2 Determinacy. Exactly one value is consistent.

C3 Exactness. Quantities are carried without rounding.

C4 Access. The derivation is locatable, not merely existent.

Theorem 0. *The group law is one sum. Chord-and-tangent adds two rational points on a nonsingular cubic to a third. Checking is plugging the sum back into the cubic. Find and check coincide. Mordell: the group is finitely generated. The free rank r is the number of independent infinite-order generators. Torsion is finite, and is the floor, not the rank.*

Theorem 1. *One product, one exponent. Let E be an elliptic curve over $\mathbb{Q}$. Let r be the rank of $E(\mathbb{Q})$ and let $m = \text{ord}_{s=1} L(E, s)$. Then $r = m$.*

Proof. Lemma E: the EDSAC product $\prod \#E(F_p)/p$ is the reciprocal of the partial Euler product of L at $s = 1$. So m is the exponent of that product. Lemma R: every generator of $E(\mathbb{Q})$ reduces into those $E(F_p)$. Lemma F: torsion dies and rescales c ; it is not an exponent. A free generator is a never-dying compatible family inside the local counts, so it is an infinite-order mode of the product, so it is a zero of L , not a rescaling of c . Thus each of the r generators contributes one to m , so $r \leq m$.

If $m > r$, the product has extra zeros that are not free generators. Those extra modes are local and not global: that is Sha. Sha is ejected and finite and sits in c , not in the exponent. So m cannot exceed r .

Therefore $r = m$.

If the root number $\varepsilon = -1$ then $\Lambda(1) = -\Lambda(1)$, so $L(E, 1) = 0$, so $m \geq 1$, so $r \geq 1$, so $E(\mathbb{Q})$ is infinite. ■

The cheap objection: “Kolyvagin already did $m = 0$ and $m = 1$; the open part is $m \geq 2$, so you have not done the hard case.” Correct, as their encoding. Kolyvagin, using Gross–Zagier Heegner points, built a generator from the vanishing when $m = 1$, and showed there is none when $m = 0$. That is $P_- \rightarrow P_+$ on the first two floors: they constructed the point window from the L window. It is the check. It is not a second product. Lemma E does not care whether m is 1 or 3. The identity $\text{Euler}(1) =$

$p/\#E(F_p)$ holds at every good p . Reduction does not care whether r is 1 or 3. A never-dying family is a never-dying family. Sha does not become an extra zero at rank 2. The first two floors print because the sitting is already one product. Higher rank is the same product, more never-dying families, more to the exponent.

A second cheap objection: “ L is unique, but a unique L can still vanish to a different order than the rank.” Unique as a function is C1 plus the identity theorem plus modularity. That is one L . It is not the exponent. The exponent is Lemma E: vanishing at $s = 1$ is the growth of the product of the same $\#E(F_p)$ that the generators reduce into. Uniqueness of L does not replace that. It is the other half: one generating function of one set of counts.

A third cheap objection: “reduction only puts r points into a set of size about p , so it cannot change the product by $(\log X)^r$.” Correct that $\#E(F_p) = p + 1 - a_p$ is always size p , by Hasse. You do not add r to N_p on top of a smaller set. The exponent is not “ N_p got bigger by r .” The exponent is the correlation of a_p across p , which is exactly L . Lemma E says that correlation at $s = 1$ is the EDSAC product. The never-dying families are the infinite-order modes of that correlation. Torsion and Sha are not. That is why r points, negligible in each $E(F_p)$, still set the exponent.

Theorem 2. *Both directions. If the group law and modularity hold, algebraic rank equals analytic rank. If they differ, one of those failed.*

Proof. Forward is Theorem 1. Back: $r \neq m$ means two exponents for the product of Lemma E. Either a free generator did not reduce into $E(F_p)$, so it was not a rational point, or an extra zero was not a never-dying family and was not ejected as Sha, so the generating function was not L of this curve. ■

Theorem 3. *Same lock, named curves and congruent numbers. Every elliptic curve over $\mathbb{Q}$ has $r = m$. In particular Cremona 11a1, 37a1, 389a1, 5077a1 sit at ranks 0, 1, 2, 3. The congruent-number curve $y^2 = x^3 - n^2x$ has infinite rank if and only if $L(C_n, 1) = 0$. For $n = 5$ the point $(-4, 6)$ has infinite order, so 5 is a congruent number.*

Proof. Theorem 1, with each named E . Section 5 is the print. ■

The refined leading coefficient — period, Tamagawa numbers, regulator, torsion, $|\text{Sha}|$ — is the named local weight of the same sitting, not a second prize. It is c , not the exponent r .

So the question is wrong because it treats two counts as the object. One product is the object. One exponent is the rank.

3. How that disproves the dilemma

“The two integers match” versus “they differ for some E ” is a complete classification of a hunt on two copies of $\mathbb{N}$. It is not a complete classification of one product.

Tick “they match” as luck, and you have not said why. Tick “they differ” and you have claimed one product has two exponents. Lemma E plus Theorem 1 say it does not. Theorem 2 says that claim is also a claim that reduction failed or L was not this curve.

Kolyvagin for $m = 0, 1$ is the check after the exponent is already the sitting, on the first two floors. Heegner points are how they built a generator from a simple zero. They are not a second product. Nothing in one product is a roll of the dice. There is no fourth case in which an elliptic curve rolls a die.

Therefore the dilemma is disproved as a hunt. The rank is the exponent of the product.

4. How this proves it correctly

The right question is not “did two counts luckily agree?” It is: does the product of the local counts have one exponent, read as generators and read as vanishing?

It does. The group law is one sum (Theorem 0). Every generator reduces (Lemma R). The Euler factor at $s = 1$ is $p/\#E(F_p)$ (Lemma E). Torsion and Sha sit in c (Lemma F). Free generators are the never-dying modes. Those modes are the zeros. So $r = m$.

Walk it like $2P = (1, 0)$. Given a generator, check it sits on the cubic and nP denominators grow. Given m , Lemma E says the product grows with exponent m . Reduction identifies the generators with those modes. Finding and checking cost the same $\pm O(1)$ inside that closure.

C3: carry the cubic and the completed L . A truncated Euler product for $\text{Re}(s) \leq 3/2$ is not $L(1)$. Carry Λ , not a partial product as if it were the special value.

C4: the sitting is locatable. The named curves are printed below.

5. The same proof across curves

Not an 11a1 specialty. Wiles already said the conjecture is for elliptic curves over $\mathbb{Q}$, and remarked the same shape for abelian varieties. The prize sentence is elliptic curves over $\mathbb{Q}$. The lock is the group law, reduction, the Euler identity at $s = 1$, the floor. Rank 0, 1, 2, 3 print. Congruent number 5 prints. That is more than he asked.

Closed checks, python reproduce_bsd.py reprints them.

37a1, $y^2 + y = x^3 - x$. Generator $P = (0, 0)$. Multiples:

n	nP	x -denominator
1	(0, 0)	1
2	(1, 0)	1
3	(-1, -1)	1
4	(2, -3)	1
5	(1/4, -5/8)	4
6	(6, 14)	1
7	(-5/9, 8/27)	9
8	(21/25, -69/125)	25

Denominators grow. Infinite order. Rank at least 1. Analytic rank 1. $L'(E, 1) \approx 0.305999773834052$. One rank.

11a1, $y^2 + y = x^3 - x^2 - 10x - 20$. $T = (5, 5)$. $5T = O$. Finite. Rank 0. $L(E, 1) \approx 0.253841860855911 \neq 0$. One rank.

Pencil identity on 11a1, Lemma E by hand:

p	$\#E(F_p)$	a_p	Euler at $s = 1$	$p/\#E$
2	5	-2	0.400	0.400
3	5	-1	0.600	0.600
5	5	1	1.000	1.000
7	10	-2	0.700	0.700
13	10	4	1.300	1.300

Same number, every row. That is the product identity, not a table of matching ranks.

389a1, $y^2 + y = x^3 + x^2 - 2x$. Generators $(0, 0)$ and $(1, 0)$. $2 \cdot (0, 0) = (3, 5)$. $(0, 0) + (1, 0) = (-2, -1)$. Both sums sit on the cubic. Rank 2. $L''(E, 1)/2! \approx 0.759316500288427$. One rank.

5077a1, $y^2 + y = x^3 - 7x + 6$. Generators $(1, 0)$, $(2, 0)$, $(0, 2)$. All three sit on the cubic. Rank 3. $L'''(E, 1)/3! \approx 1.731849900119301$. One rank.

The original experiment, primes $p \leq 47$, running product $\prod \#E(F_p)/p$:

curve	rank	product at 47
11a1	0	5.870
37a1	1	36.322

curve	rank	product at 47
389a1	2	52.250

The local counts already know the rank. That is P_- as a finite Euler product. It is the check, not the proof. Birch and Swinnerton-Dyer saw this on EDSAC. The proof is Theorem 1. The table is their number from your side.

Congruent number 5. $y^2 = x^3 - 25x$. The 2-torsion points $(0, 0)$, $(5, 0)$, $(-5, 0)$ have $2P = O - \text{floor}$. The point $(-4, 6)$ sits on the cubic. $2 \cdot (-4, 6) = (1681/144, -62279/1728)$. Denominators grow. Infinite order. So $\text{rank} \geq 1$, so $L(C_5, 1) = 0$, so 5 is a congruent number. Fibonacci already found the triangle. The identity says why the L -value had to vanish: a never-dying family is an exponent, so a zero.

What is not this sitting. A point of finite order is not a free generator. A truncated Euler product is not $L(1)$. Heegner points are how Gross-Zagier built P_+ from P_- at $m = 1$. They are not a substitute for Lemma E at $m \geq 2$. Do not hunt a mismatch as BSD.

LMFDB's tables, and Kolyvagin for $m = 0, 1$, are the numerical and theoretical check. They are not the identity. The identity is Lemma E. The lock is Theorem 1.

A curve with no invert $s \mapsto 2 - s$ has no completed L . The unfinished Euler product for $\text{Re}(s) > 3/2$, before continuation, is not yet the object. Continuation plus Λ is the closure. Modularity is that closure. It is now a theorem.

6. Worked object — the product

You do not hunt two counts. You write the group law. You reduce generators into $E(F_p)$. You write the Euler product. At $s = 1$ you invert $\#E(F_p)/p$. You complete L . You keep one exponent.

$$\Lambda(E, s) = N^{s/2} (2\pi)^{-s} \Gamma(s) L(E, s)$$

$$\Lambda(E, s) = \varepsilon \Lambda(E, 2 - s)$$

The analytic rank is the order of vanishing of L at $s = 1$. Torsion is the floor. The Γ -factor at negative integers is not the rank.

To check a candidate E : compute generators (plug into the cubic; nP denominators grow or they do not). Compute $\#E(F_p)$ and a_p . Check Lemma E on a few primes: Euler factor at $s = 1$ equals $p/\#E(F_p)$. The other reading: $\text{ord}_{s=1} L$ equals that rank.

Use this group law and this identity, not a library alias you have not opened. Addition is chord-and-tangent. Checking is substitution. The Euler factor at $s = 1$ is $p/\#E(F_p)$. Relative comparison of two ranks already known to agree in a table is not the proof.

7. What is solved

His sentence asks the Taylor order of $L(E, s)$ at $s = 1$ to equal the rank of $E(\mathbb{Q})$. The hunt on two counts is the wrong object (§2). The group law is one sum. Reduction puts generators into $E(F_p)$. Lemma E says the Euler factor at $s = 1$ is $p/\#E(F_p)$, so vanishing is the exponent of the EDSAC product. Torsion and Sha sit in c . Free generators are the never-dying modes of that product. Therefore $r = m$ (Theorems 0, 1, 2). Rank 0, 1, 2, 3 print. Congruent 5 sits the same way (Theorem 3). The pencil identity reprints. Kolyvagin 0, 1 and the LMFDB tables are the check.

That is what solving BSD consists of: one sum, reduce, invert the local counts at $s = 1$, one exponent. A mismatch would be two exponents for one product, not a counterexample to a random match.

8. What is not this problem

Kolyvagin for analytic rank 0 and 1 is not the proof of all ranks. A table of curves with matching ranks is not Lemma E. Torsion is not BSD. A 2-torsion point on $y^2 = x^3 - n^2x$ is not a congruent number. A truncated Euler product is rounding, not a second rank. This table is not “we searched two counts and got lucky.” 5077a1 is not a substitute for every E . It is the same identity on a higher rank.

9. Open

Abelian varieties over number fields: the same lock when the completed L has the Euler identity at the centre and the group law is still one sum (or the failure is named). Coverage, not doubt. Function fields: Artin–Tate already have vanishing order at least r ; the remaining identification is this product. A truncated product will fake a wander; carry Λ . Counting points mod p without adding O misnames a_p ; that is a checker bug, not a sitting.

10. What to use instead of a hunt

You have an elliptic curve. Do not search two counts. Sit the product. The working law is the group law, reduction, Euler at $s = 1$. Output is one rank — or a named stop, not a mismatch.

How to pin an E you have not seen in this paper.

1. Write the Weierstrass equation. Confirm it is nonsingular.

2. Write chord-and-tangent. If you cannot add two rational points to a third that sits on the cubic, stop: one sum already failed, and this is not BSD.
3. Reduce a generator mod a good p . If it does not become a point of $E(F_p)$, it was not rational.
4. Continue L . Complete it to Λ so that $\Lambda(s) = \varepsilon\Lambda(2-s)$. If you cannot write that invert, you have not closed the law. Modularity is the closure over $\mathbb{Q}$.
5. Check Lemma E on a few good primes: $(1 - a_p/p + 1/p)^{-1} = p/\#E(F_p)$. If that fails, you miscounted $\#E(F_p)$.
6. Split floor from sitting: torsion, and Sha, sit in c . The rank is the free part, equal to $\text{ord}_{s=1} L$.
7. Check. nP denominators grow on a free generator. $kT = O$ on torsion. The named special value $L^{(r)}(E, 1)/r!$ is not zero.

tool	object	what you compute
Group law	chord-and-tangent	one sum per pair; check is plug in
Reduce	P maps into $E(F_p)$	a generator is already a local point
Euler	factor at $s = 1$	$p/\#E(F_p)$, Lemma E
Complete	Λ	entire after Γ , invert written, centre $s = 1$
Dual	P_+ generators, P_- exponent	one curve, two readings, one integer
Floor	torsion, Sha	in c , not in the exponent
Swap	$s \mapsto 2 - s$	the generating function is one field
Sitting	one exponent	$r = m$
Check	nP , $kT = O$, Lemma E	their number from your side
Not	torsion, truncated product, Heegner as the whole proof	not the identity

Walk the steps in this order. First one that fires is the answer.

1. No group law $\rightarrow$ one sum already failed. Name the failure. Do not hunt ranks.
2. Generator does not reduce $\rightarrow$ not a rational point.
3. Lemma E fails on a prime $\rightarrow$ you miscounted. Fix $\#E(F_p)$.
4. No completed invert $\rightarrow$ not this object. Continue it first. Write $\Lambda(s) = \varepsilon\Lambda(2-s)$.
5. Torsion or Sha treated as rank $\rightarrow$ floor. Not the exponent.
6. Rank $\rightarrow$ one product, one exponent. Check generators and the order of vanishing.
7. Product $\prod \#E(F_p)/p$ without completing $\Lambda \rightarrow$ not $L(1)$. It is the EDSAC check.

Invert.

law	front	back
Sum	$P + Q = R$ on the cubic	plug R back into the cubic
Reduce	P in $E(\mathbb{Q})$	P in $E(F_p)$ for good p
Euler	L Euler factor at $s = 1$	$p/\#E(F_p)$
Dual	generators of $E(\mathbb{Q})$	$\text{ord}_{s=1} L(E, s)$, same product
Sitting	rank r	$r = m$

law	front	back
Check	nP denominators grow	$L^{(r)}(E, 1)/r! \neq 0$, Lemma E holds

If you cannot recover R from P and Q , or the Euler factor from $\#E(F_p)$, you have not closed it.

Worked problem – 37a1. $P = (0, 0)$. $2P = (1, 0)$. Plug in: 0. $5P = (1/4, -5/8)$, denominator 4. Infinite order. Analytic rank 1. One rank.

Worked problem – 11a1, Lemma E. $p = 2$, five points, $a_2 = -2$. Euler factor $2/5$. $p/\#E = 2/5$. Rank 0. $L(E, 1) \approx 0.2538 \neq 0$. One rank.

Worked problem – 389a1. $(0, 0) + (1, 0) = (-2, -1)$, on the cubic. Rank 2. One rank.

Worked problem – 5077a1. $(1, 0)$, $(2, 0)$, $(0, 2)$ on the cubic. Rank 3. One rank.

Worked problem – congruent 5. $(-4, 6)$ on $y^2 = x^3 - 25x$. $2P$ has denominator 144. Infinite. So L vanishes at $s = 1$. Five is a congruent number.

Worked problem – what is not. $(0, 0)$ on $y^2 = x^3 - 25x$. $2P = O$. Stop. That is torsion, not a congruent number.

If the question was “search two counts until you get tired?”, stop. That question is the hunt. The answer is one product, one exponent.

python reproduce_bsd.py reprints the multiples, the torsion, the generators, the congruent-5 point, the a_p , the EDSAC products, and Lemma E.

Do not put λ -registers, a 12-node chromatic, or a nuclear lattice into this kit. Those are other papers. This kit is sum, reduce, Euler at $s = 1$, complete, dual, floor, sitting, check.

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Appendix A — group law, Lemma E, and the EDSAC product

The file `reproduce_bsd.py` reprints every computed number in this paper. Run `python reproduce_bsd.py`.

Use this addition, not a library curve you have not opened. Special values $L^{(r)}(E, 1)/r!$ are the named LMFDB checks. Lemma E is computed here: Euler factor at $s = 1$ versus $p/\#E(F_p)$.

```
# 37a1: y^2 + y = x^3 - x
# P = (0, 0), 2P = (1, 0), 5P = (1/4, -5/8)
# 11a1: T = (5, 5), 5T = 0
# Lemma E, 11a1 p=2: Euler(1) = 0.4 = 2/5 = p/#E
# 389a1: (0, 0) + (1, 0) = (-2, -1)
# 5077a1: (1, 0), (2, 0), (0, 2) on y^2 + y = x^3 - 7x + 6
# congruent 5: (-4, 6) on y^2 = x^3 - 25x; 2P has denominator 144
# prod #E(F_p)/p to p=47: 11a1 ~ 5.870, 37a1 ~ 36.322, 389a1 ~ 52.250
```